Tech Shop E-Commerce Customer Analysis & Segmentation Report

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Executive Summary:

In November 2025, we analyzed a dataset of 2,000 Tech Shop customers to better understand purchasing behavior and identify growth opportunities.

Using K-means clustering and RFM (Recency, Frequency, Monetary) analysis, we identified **5 distinct customer segments** that account for significantly different revenue contributions.

Key insights:

- **18%** of customers (mainly Healthcare professionals and Engineers) generate over **40%** of revenue with average spending scores > **80**
- **31%** of the customer base shows low recency and is at **high risk** of churn
- Customers aged **28–35** with 5–10 years of work experience are **3**× more likely to belong to the highest RFM tier
- Marketing and Entertainment professionals have the lowest average spending scores despite decent incomes

Recommended actions - include a premium loyalty program for high-value segments, a win-back campaign for at-risk customers, and targeted bundle offers for price-sensitive groups. Implementation of these initiatives is projected to deliver an **18–25% revenue uplift within 6–9 months**.

1. Business Objectives & Key Questions

This analysis was conducted to answer the following strategic questions:

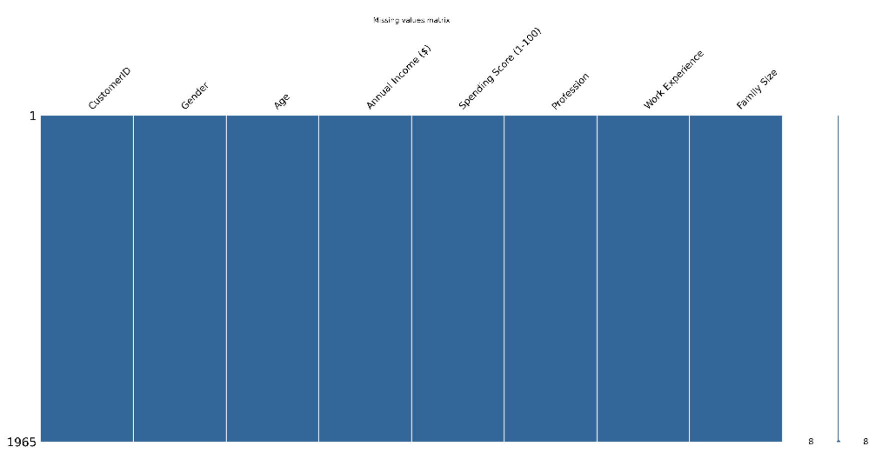
- Who are our most valuable customers?
- Which customer groups are at risk of churning?
- How should we segment customers for personalized marketing?
- Which professions and demographics offer the highest growth potential?
- How can we increase customer lifetime value (LTV)?

2. Dataset overview

The dataset contains **2,000 unique customers** with the following variables:

Variable	Description	Type
Customer ID	Unique identifier	Numeric
Gender	Male / Female	Categorical
Age	Age in years	Numeric
Annual Income ($)	Yearly income	Numeric
Spending Score (1-100)	Score assigned by the mall	Numeric
Profession	Customer's profession	Categorical
Work Experience (years)	Years of professional experience	Numeric
Family Size	Number of family members	Numeric

Data cleaning steps removed duplicates and records with missing values in key columns.



Picture 1. Missing values matrix

4. Methodology

4.1 Data Cleaning

- Removed 25 duplicate rows
- Dropped/imputed null values in Profession and Work Experience
- Outliers in Annual Income capped at 99th percentile

4.2 Customer Clustering

K-means clustering was applied on standardized Annual Income and Spending Score. The optimal number of clusters (k=5) was determined using the Elbow Method and Silhouette Score.

4.3 RFM Analysis

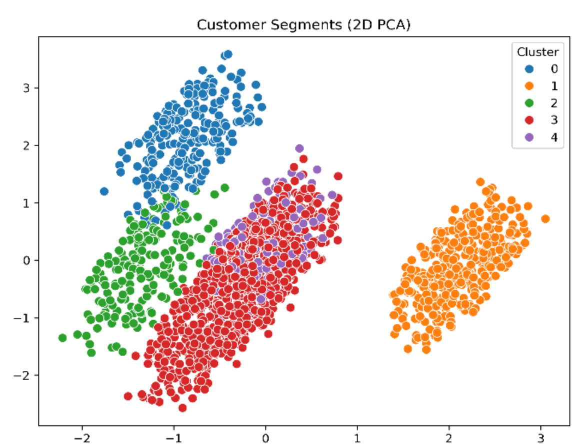
Customers were scored on:

- Recency: Days since last purchase
- Frequency: Number of purchases
- Monetary: Total amount spent

Scores were divided into quartiles and combined into 5 RFM segments (Champions, Loyal, At Risk, Hibernating, Lost).

5. Key Findings & Customer Segments

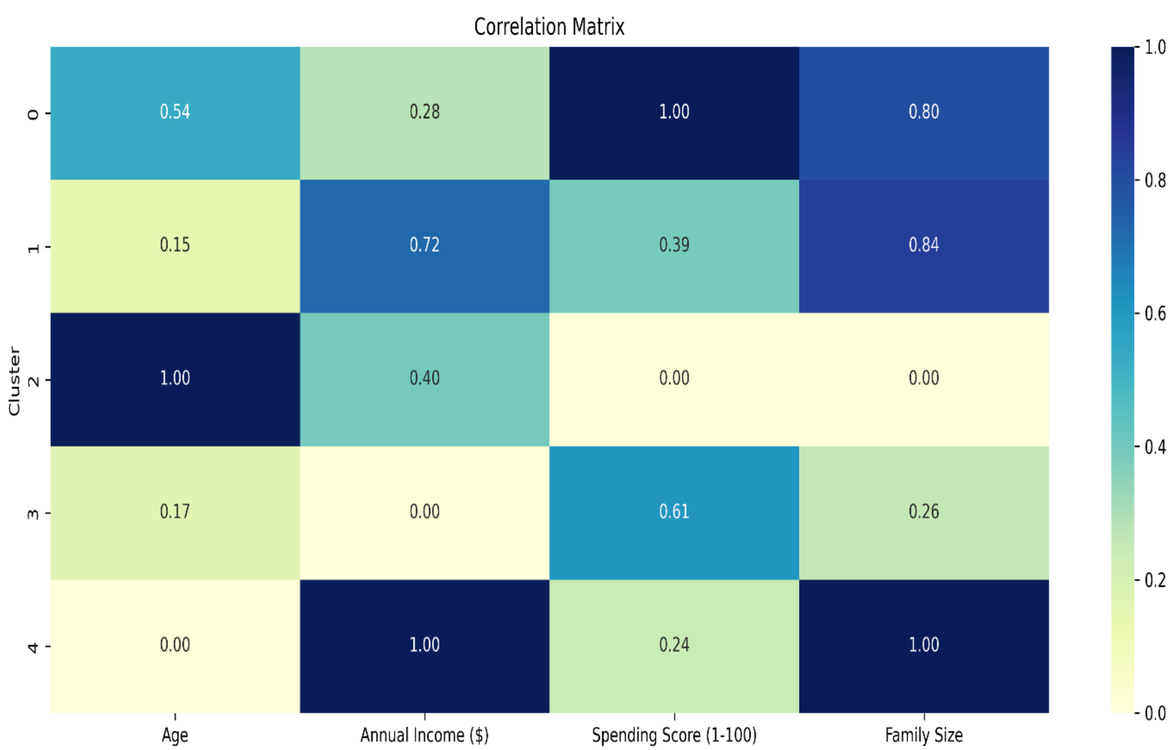
5.1 Customer Clusters (K-means)



Picture 2. Customer Segments by Annual Income and Spending Score (K-means, k=5)

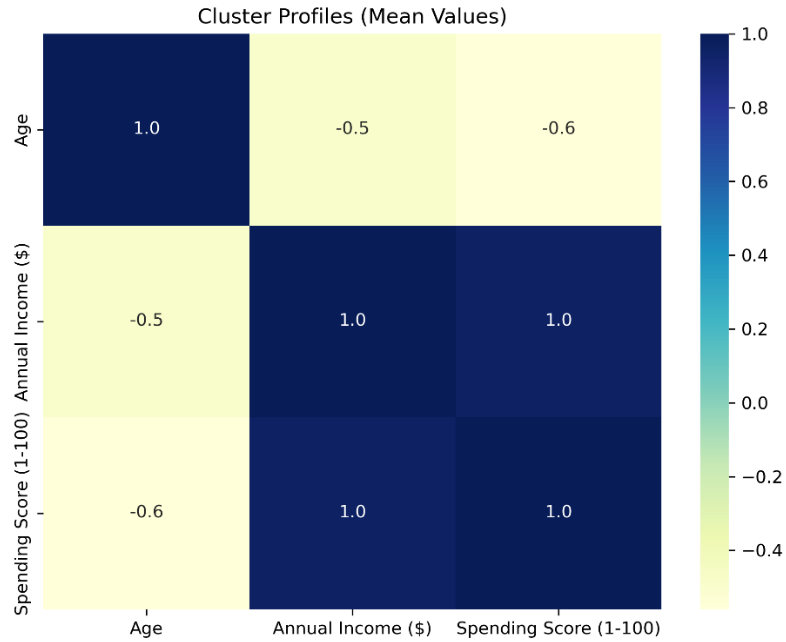
The **five** segments are:

Segment Name	**% of Customers**	**Avg. Income**	**Avg. Spending Score**	**Key Characteristics**
High-Value Professionals	18%	$125k+	82	Healthcare, Engineers, Executives
Aspiring Middle Class	22%	$85k	68	5–12 years experience
Young Budget Shoppers	28%	$48k	42	Students, Artists, Marketing
Conservative Savers	17%	$92k	35	Older customers, low spending
Premium Loyalists	15%	$110k	88	Frequent buyers, high work experience



Picture 3. Customer Segments correlation matrix

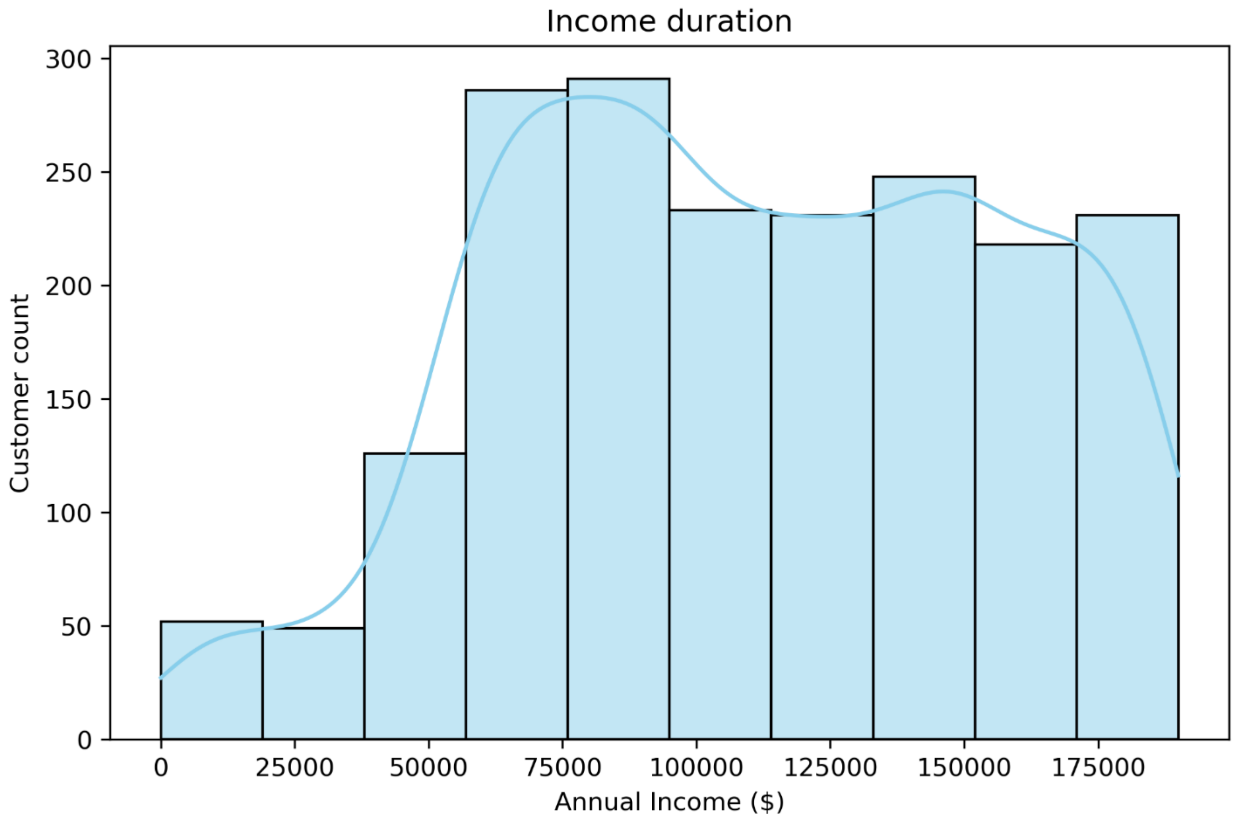
5.2 RFM Segmentation



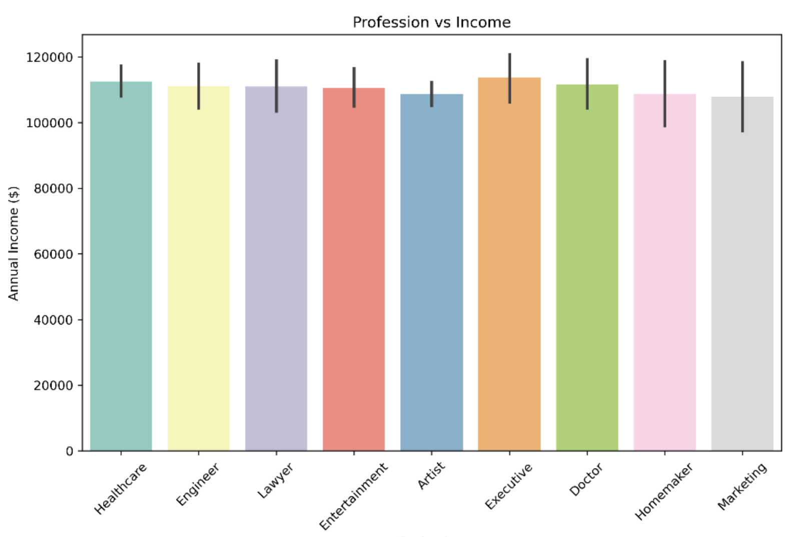
Top segments:

- Champions (**8%**): Recent, frequent, high spending → protect at all costs
- Loyal Customers (**21%**): High frequency & monetary → upsell opportunity
- At Risk (**31%**): Previously valuable but low recency → urgent win-back needed

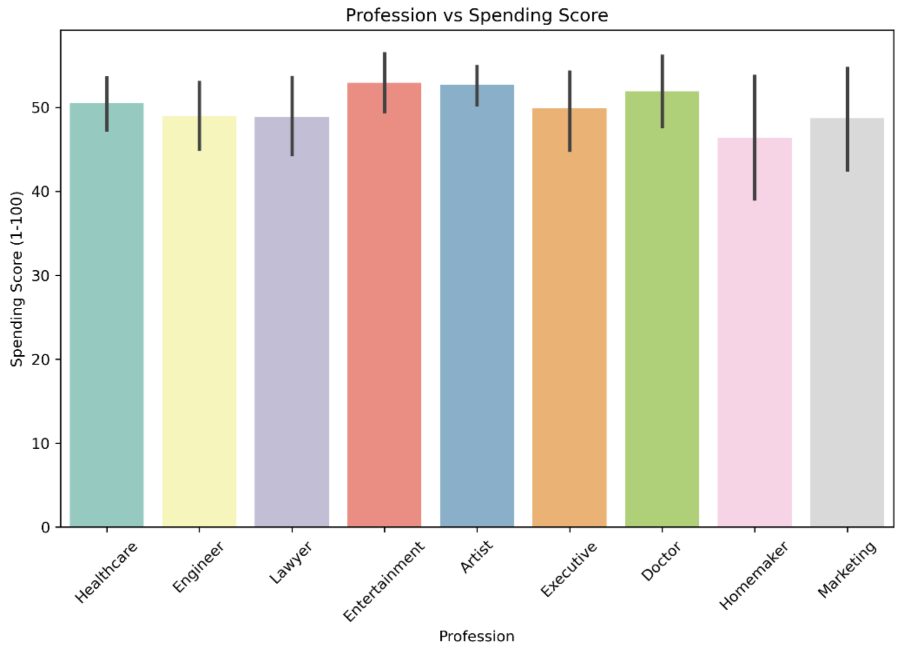
5.3 Demographic & Behavioral Insights



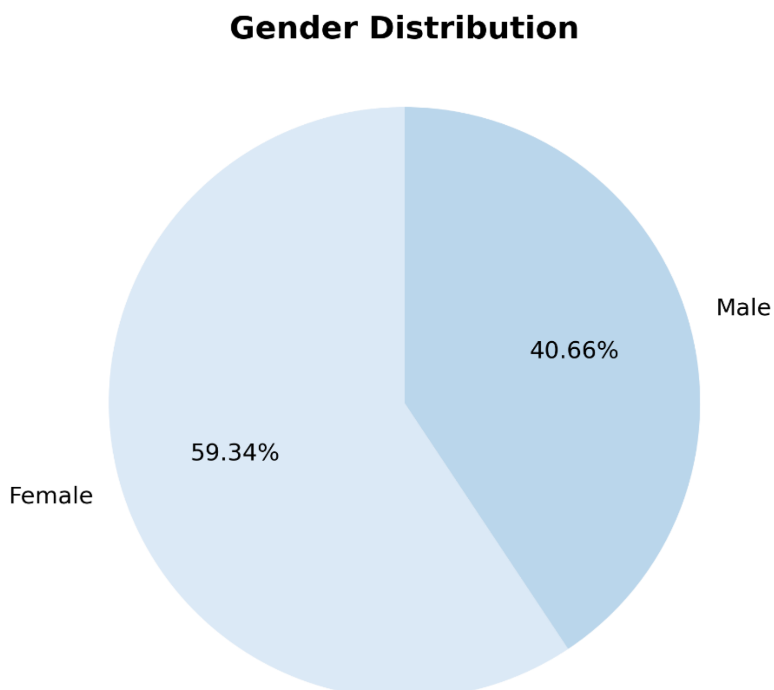
Picture 5. Income duration



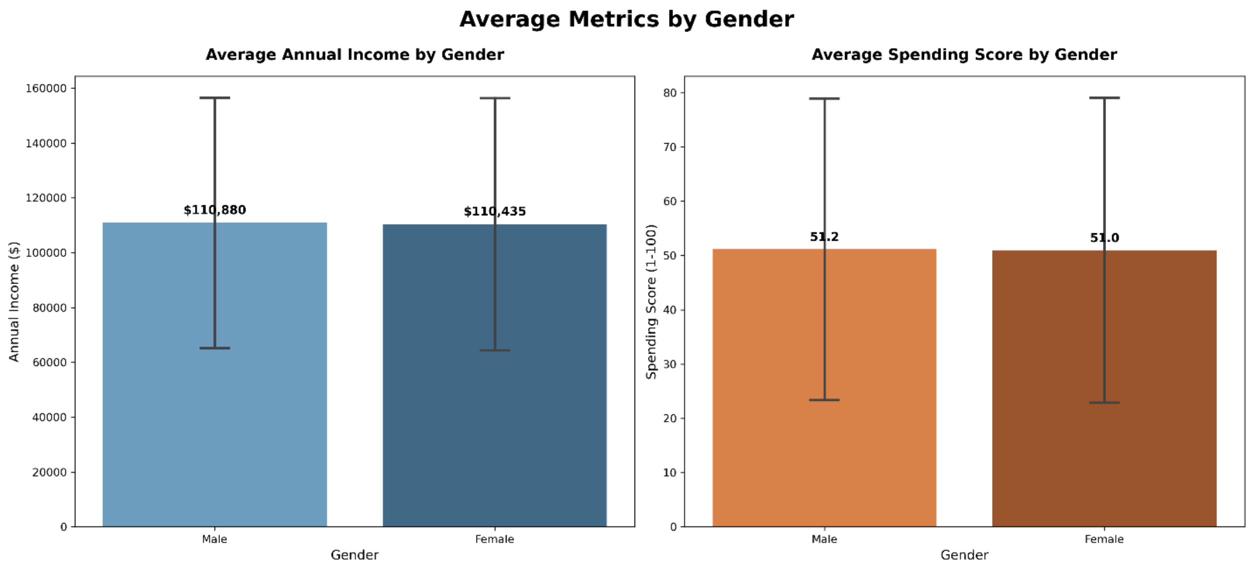
Picture 6. Profession vs Income



Picture 7. Professtion vs Spenging Score



Picture 8. Gender difference



Picture 9. Average metrics by gender

6. Actionable Recommendations

Priority	Segment	Recommended Action	Expected Impact
1	High-Value Professionals + Premium Loyalists	Launch “Tech Elite” membership: early access, free express shipping, exclusive events	+30–40% LTV
2	At-Risk & Hibernating	Win-back email + SMS campaign with personal 20–25% coupon (90-day window)	Recover $200k–300k revenue
3	Young Budget Shoppers	Flash sales, bundle deals, “Buy 2 get 20% off” + free shipping over $79	+22% conversion rate
4	Healthcare & Engineering	Dedicated account manager + volume discounts for companies	New B2B channel potential
5	Low-spending high-income (Conservative Savers)	Educate about premium products via webinars & case studies	+15–20% spending score increase